

1 • Training only: evidence habits

The spring science evidence exchange

Name _____ Date _____

Use during I do and We do. Keep the checkpoint route face down until You do. The numbers below are constructed toy-car data.

Surface	Trial 1 / cm	Trial 2 / cm
A	60	58
B	40	42

Same method

Same toy car, ramp height, release point and distance rule. Only the surface changes.

Evidence frame

In these trials, _____. For example, _____ cm compared with _____ cm.

Claim to improve

“The car always travels farther on A in every situation.” Two trials cannot establish every possible condition.

Access

Questions may be read aloud. A scribe records your own explanation, not an adult's answer.

W1. On which surface did the car travel farther in both trials?

W2. Give one pair of values and units.

W3. Improve the claim and suggest one useful next check.

2 • Supported checkpoint: three science cases

Open only at You do. Work independently. Teacher may read the words.

Case A: a ladybird has egg → larva → pupa → adult stages. A common frog has egg → tadpole → froglet → adult stages.

S1. Which animal has a pupa stage?

- A. Ladybird
- B. Common frog

S2. Give one life-cycle similarity and one difference.

S3. Which job belongs to the heart?

- A. Pumps blood
- B. Digests whole food

Case B: constructed cooling data. Same bottles, starting at 30 °C, in a 20 °C room. After 10 minutes: no sleeve 24 °C; thick sleeve 28 °C.

S4. Which bottle cooled less in this test?

- A. No sleeve
- B. Thick sleeve

S5. Explain your choice using one temperature. Does the sleeve stop all energy transfer forever?

3 • Standard checkpoint: explain the evidence

Open only at You do. These are new cases and constructed data.

Life-cycle cards: ladybird - egg, larva, pupa, adult; dragonfly - egg, aquatic nymph, adult. Both adults can produce a new generation.

N1. Give one similarity and one difference between these cycles. Explain the arrow from adult to new eggs.

N2. Explain how digestion and blood transport work together. State the heart's job.

Bottle sleeve	Start / °C	After 10 min / °C
None	30	24
Thick fabric	30	28

Constructed sample data: same bottles and fill, same 20 °C room, one trial per sleeve.

N3. Which bottle cooled less? Calculate both falls and explain the effect of insulation.

N4. Give one limit and one useful improvement to this test.

4 • Stretch checkpoint: test the claims

Open only at You do. Support each explanation; do not claim more than the cases show.

Life-cycle evidence: ladybird has egg → larva → pupa → adult. Dragonfly has egg → aquatic nymph → adult. Platypus is an egg-laying mammal whose young receive milk.

T1. Evaluate “every young insect becomes a pupa” and “all egg-layers are birds”. Use a named counterexample for each.

T2. Repair “The heart digests food and sends whole crumbs to tissues”. Link two systems accurately.

Sleeve	Start / °C	10 min / °C	Thickness
None	30	24	0 mm
Fabric	30	28	8 mm

Constructed data: same bottles and fill; 20 °C room; one trial each.

T3. Compare temperature falls. Explain why this does not prove fabric stops all energy transfer.

T4. Design one additional comparison to test thickness. Name a control and the measurement.

5 • Feedback and next step

Complete after submitting the independent checkpoint. This page records improvement, not a new independent score.

E1. Which area is most secure: life cycles, body systems or materials? Name evidence from your answer.

E2. Write one corrected scientific explanation after feedback.

E3. What observation, number or process supports the correction?

E4. Choose one precise next practice task.

Reflection. Which kind of support helped you show your own science?
